

The minimum element of D doesn't necessarily correspond to neighbors
However it can be used as a heuristic

In a rooted tree, the root has degree 2

UPGMA:

First put every present-day species as its own cluster

Take minimum distance and form a cluster

Recompute average distance to clusters

However this doesn't fit a tree to a matrix

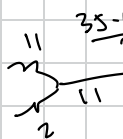
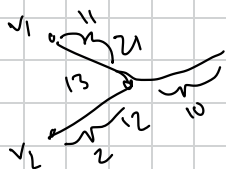
↳ We have to do a weighted average

Given an $n \times n$ matrix D its neighbor-joining matrix D^* is

$$D^*_{ij} = (n-2) \cdot D_{ij} - \text{TotalDistance}_D(i) - \text{TotalDistance}_D(j)$$

Neighbor-Joining Theorem: D additive $\Rightarrow$ smallest element of D^* corresponds to neighboring elements.

Join two neighbors and average the link length from every choice of the third node k



$$\frac{35-13}{2} = 11$$

0	10	11	21		-34-34
10	0	13	23	$\Rightarrow$	-34
11	13	0	24		

Nothing is said about ancestral states

Furthermore, information is lost when converted to a distance matrix

The Small Parsimony Algorithm

Given an assignment of strings to ancestral nodes,
how do we know how good it is?

Parsimony Score: Sum of Hamming distance along each edge

Parsimony - Occam's Razor

Given a set of leaves with strings of length m on a rooted tree,

We want to find the strings of internal nodes that minimizes the Parsimony score.

We actually can just do it symbol by symbol as they are independent w/ regards to Parsimony Score

We can do dp on subtree